

MERIDIAN INSTITUTE OF TECHNOLOGY • FALL 2026

CS 3410: Advanced Algorithms

Problem Set 4: Optimization, Flow & Amortization

Department of Computer Science & Electrical Engineering

SCORE / EVALUATION

100 / 100

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Academic Integrity Statement: All solutions contained herein represent original work conducted in accordance with Meridian Institute of Technology's Code of Academic Integrity. Computational code, proofs, and flow graphs were derived independently.

Problem 1: Amortized Complexity & Dynamic Arrays**[25 Marks]**

Consider a dynamic array data structure that expands when full by a growth factor of $\alpha = 1.5$.

- a) [15 Marks] Use the **Potential Method** to prove that insertion is $O(1)$ amortized. Define potential function $\Phi(D_i)$ and derive the exact upper bound.
- b) [10 Marks] Given an edge (u, v) with weight w_{new} added to an MST T , describe an $O(|V|)$ update algorithm and prove correctness via the Cycle Property.

1. Definition of State and Potential Function: Let k_i be elements stored after operation i , and S_i be capacity. Resizing triggers when $k_{i-1} = S_{i-1}$, setting $S_i = 1.5S_{i-1}$. We define potential function $\Phi(D_i)$:

$$\Phi(D_i) = C \cdot \left(k_i - \frac{S_i}{1.5} \right) \quad (1)$$

- After resize: $k_i = \frac{S_i}{1.5} + 1 \implies \Phi(D_i) = C > 0$.
- Full array ($k_i = S_i$): $\Phi(D_i) = C \left(S_i - \frac{S_i}{1.5} \right) = \frac{C}{3} S_i$. Thus $\Phi(D_i) \geq \Phi(D_0) = 0$.

2. Amortized Cost Derivation ($\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$):

- No Resize ($k_{i-1} < S_{i-1}$): $c_i = 1$, $\Delta\Phi = C(1) \implies \hat{c}_i = 1 + C$.
- Resize ($k_{i-1} = S_{i-1}$): $c_i = S_{i-1} + 1$, $S_i = 1.5S_{i-1}$.

$$\begin{aligned} \Delta\Phi &= C \left((S_{i-1} + 1) - \frac{1.5S_{i-1}}{1.5} \right) - C \left(S_{i-1} - \frac{S_{i-1}}{1.5} \right) = C - \frac{C}{3} S_{i-1} \\ \hat{c}_i &= (S_{i-1} + 1) + C - \frac{C}{3} S_{i-1} = 1 + C + \left(1 - \frac{C}{3} \right) S_{i-1} \end{aligned}$$

Choosing $C = 3$ cancels S_{i-1} , yielding fixed amortized bound $\hat{c}_i = 1 + 3 = 4 = O(1)$. ■